

CC: 2.1.218
OrchestKit: 8.84.4
Audit Date: 2026-07-23

OrchestKit Audit: CC 2.1.215-218 Adoption & Orchestration Correction

Technical Deep-Dive & Remediation Plan for Engineers

Core Thesis: ork encodes an orchestration model CC no longer implements.

The frontmatter is a contract with the scheduler. When defaults move, the unstated defaults silently change meaning. These are not syntax errors; they are silent scheduling relocations.

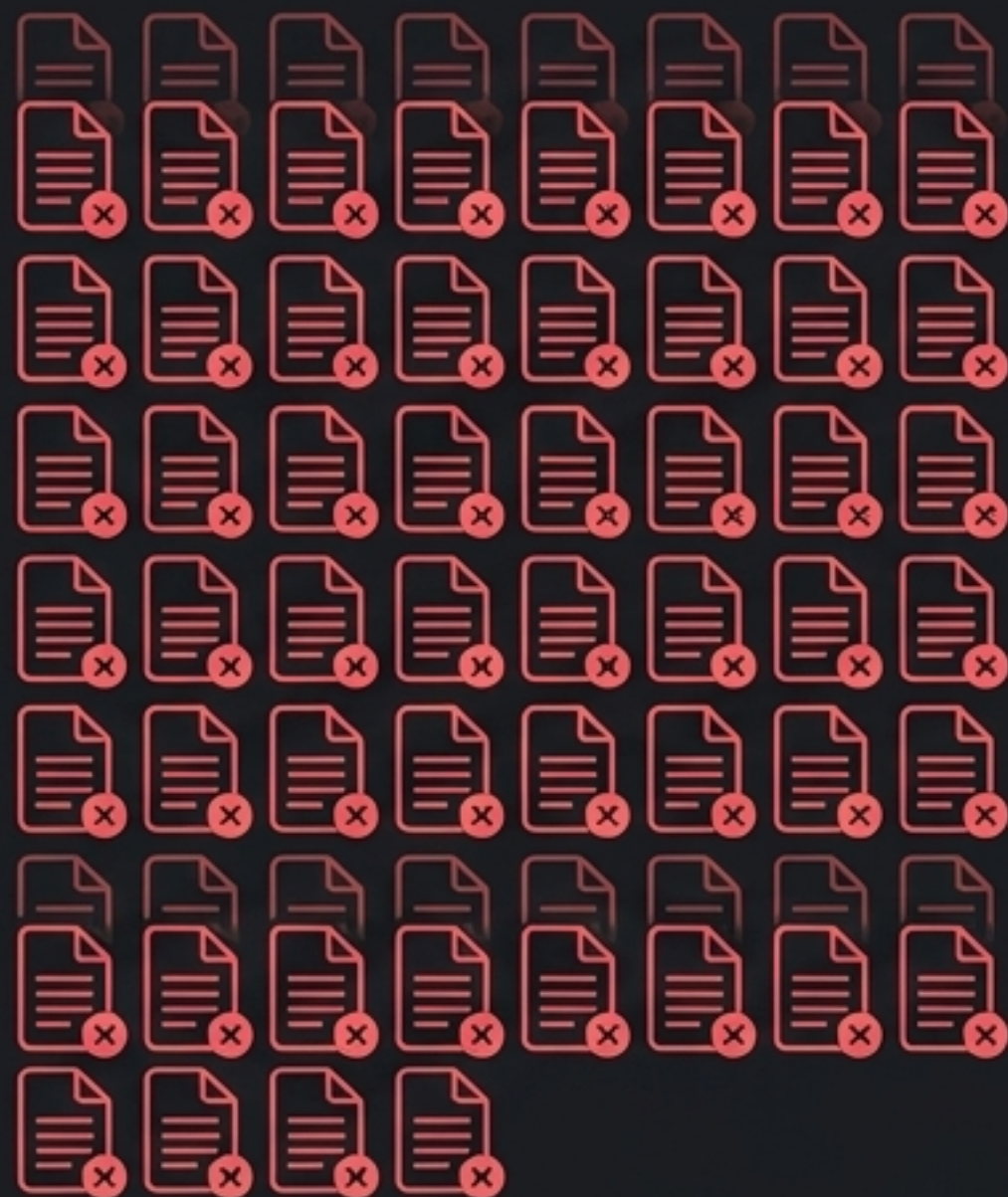
	Implicit Contract		Runtime Reality
01	<pre>frontmatter { scheduler: 'ork.default.sched.v1', // Expected static configuration defaults: { priority: 50, affinity: 'soft' } }</pre>	01	<pre>frontmatter { scheduler: 'cc.dynamic.sched.v2', // Actual runtime resolution defaults: { priority: 'omitted', affinity: 'implicit' } }</pre>
10	ork.default.sched.v1 Apply Priority: 50 Apply Affinity: 'soft' <pre>schedule(task_id, defaults); // -> task scheduled with priority 50, soft affinity</pre>	10	cc.dynamic.sched.v2 Dynamic Priority: >100 (calculated) Hard Affinity (calculated) <pre>schedule(task_id, resolved_defaults); // -> task scheduled with dynamic priority, hard affinity</pre>
20	<pre>resource 'task_queue' { min_priority: 30, // Task with priority 50 fits max_capacity: 100 }</pre>	20	<pre>resource 'task_queue' { min_priority: 30, // Task with priority > 100 rejected max_capacity: 100 + status: 'REJECTED (Priority > 100)' }</pre>

Priority 1: Fork Skills Default to Background

78 total fork skills -> 39 slash-only
-> 39 model-invocable (High Risk)

BEFORE

78 of 114 skills declare context: fork. Zero declare background:.



MECHANISM

CC 2.1.218 changed context: fork to run in the background by default. Output (pickers, diffs) is generated where the user isn't looking.



AFTER

Every fork skill explicitly declares execution intent (**background: true** or **false**). Interactive skills pin false.

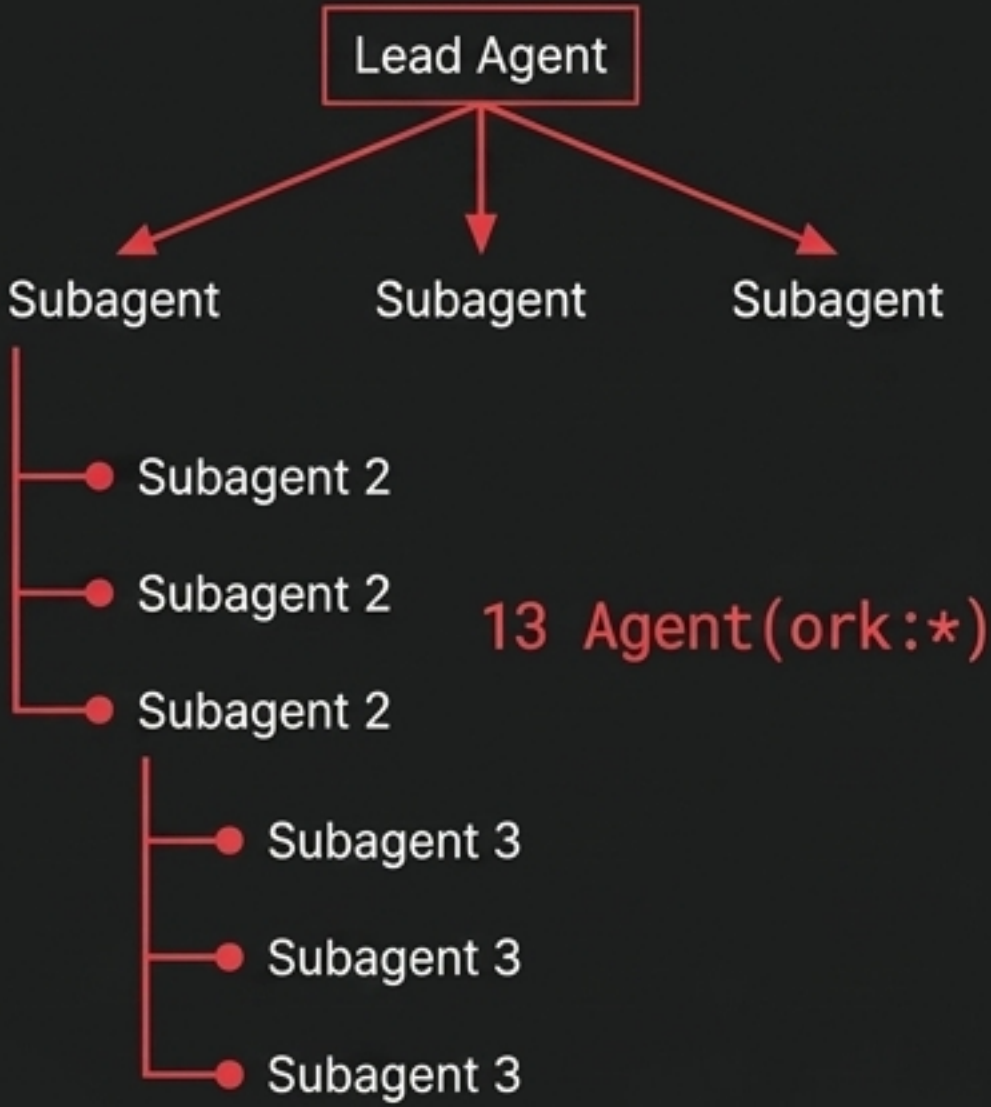
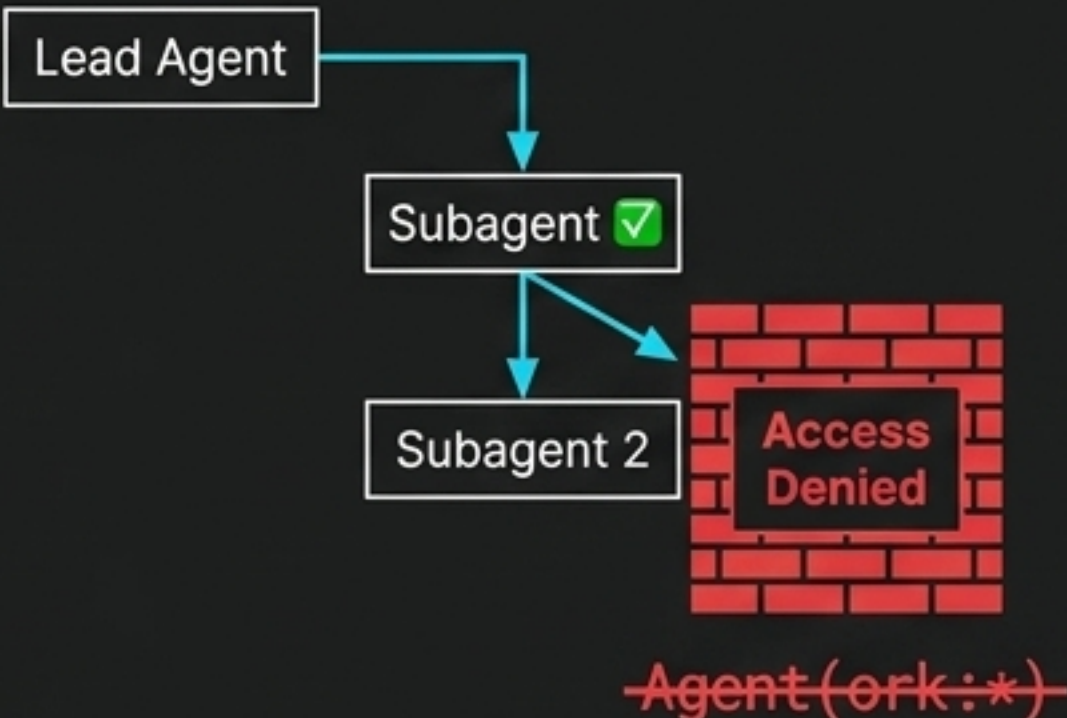
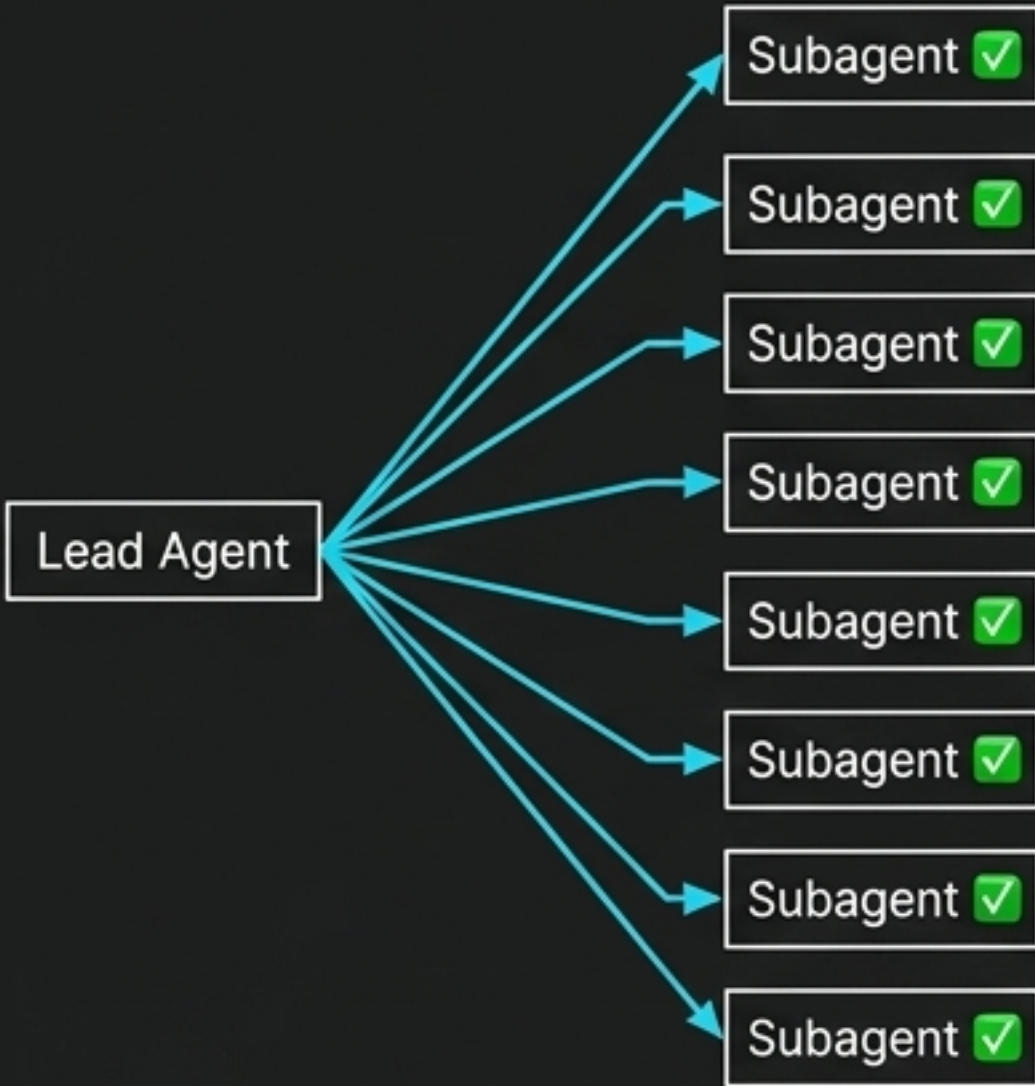
```
# Skill 1: Interactive
skill_interactive:
  context: fork
  background: false ✓

# Skill 2: Background Job
skill_background:
  context: fork
  background: true ✓

# Skill 3: Explicit Intent
skill_explicit:
  context: fork
  background: false ✓
```

Priority 2: Nested Subagent Spawning Off By Default

Telemetry:
13 inert grants | 11 impacted agents | Max subagents: 20

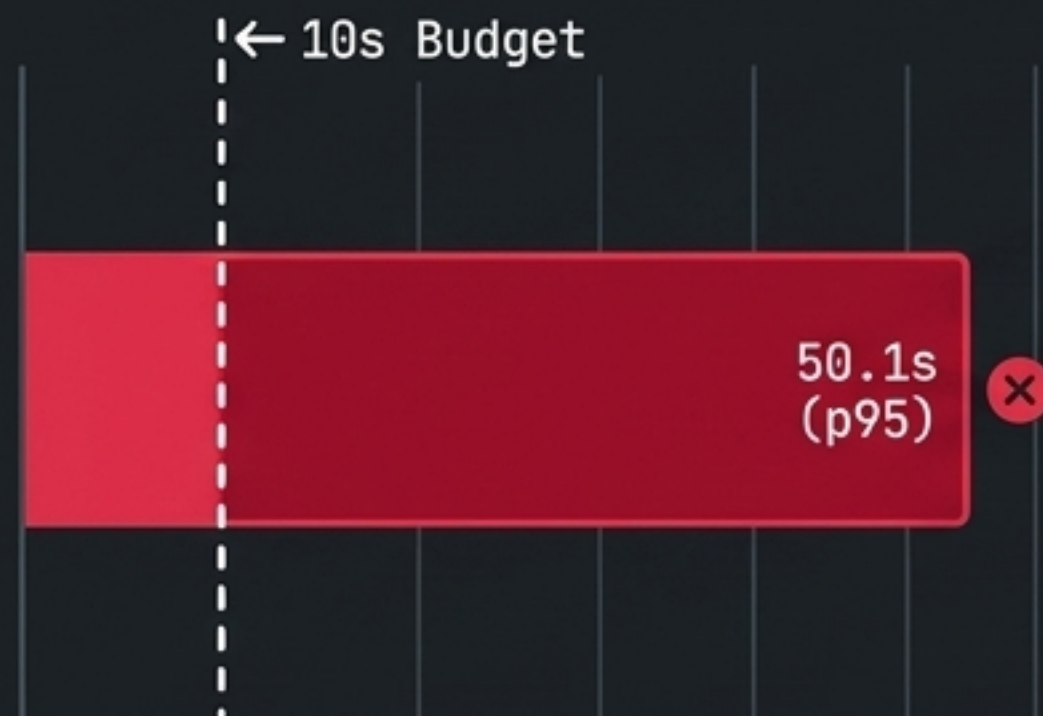
BEFORE	MECHANISM	AFTER
13 Agent(ork:*) grants across 11 agents exist. Documentation relies on 3-level deep nesting.  13 Agent(ork:*)	CC 2.1.217 requires CLAUDE_CODE_MAX_SUBAGENT_SPAWN_DEPTH to nest. A tool grant is permission, not capability. The 13 grants cost tokens but are inert, causing agents to guess inline instead of delegating. 	Own the cost explicitly via env vars, or remove grants and fan out flat from the lead agent. 

Priority 3: security-scan-aggregator Dominates Hook Time

202,229 records | 87% hook
wall time | Max: 159.6s

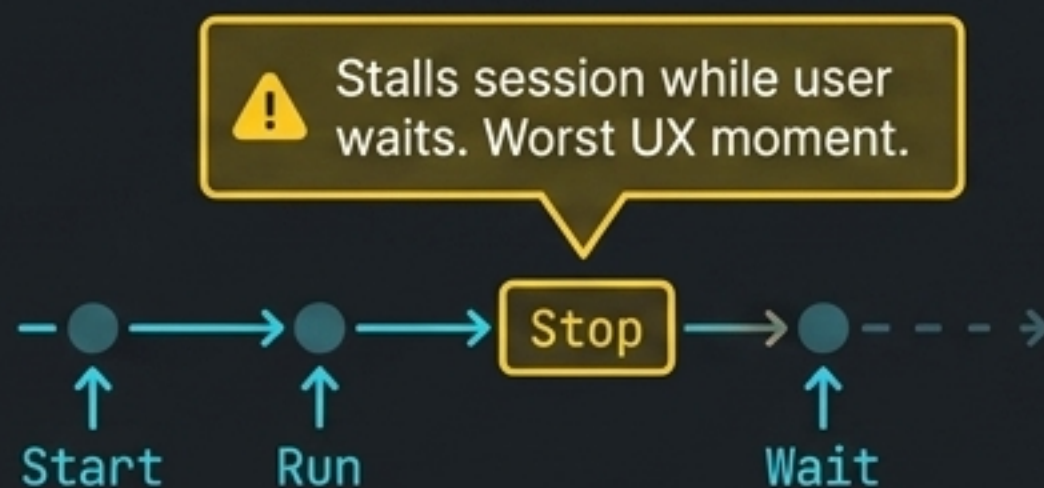
BEFORE

stop/security-scan-aggregator
consumes 87% of all hook wall time.
Declared budget: 10s.



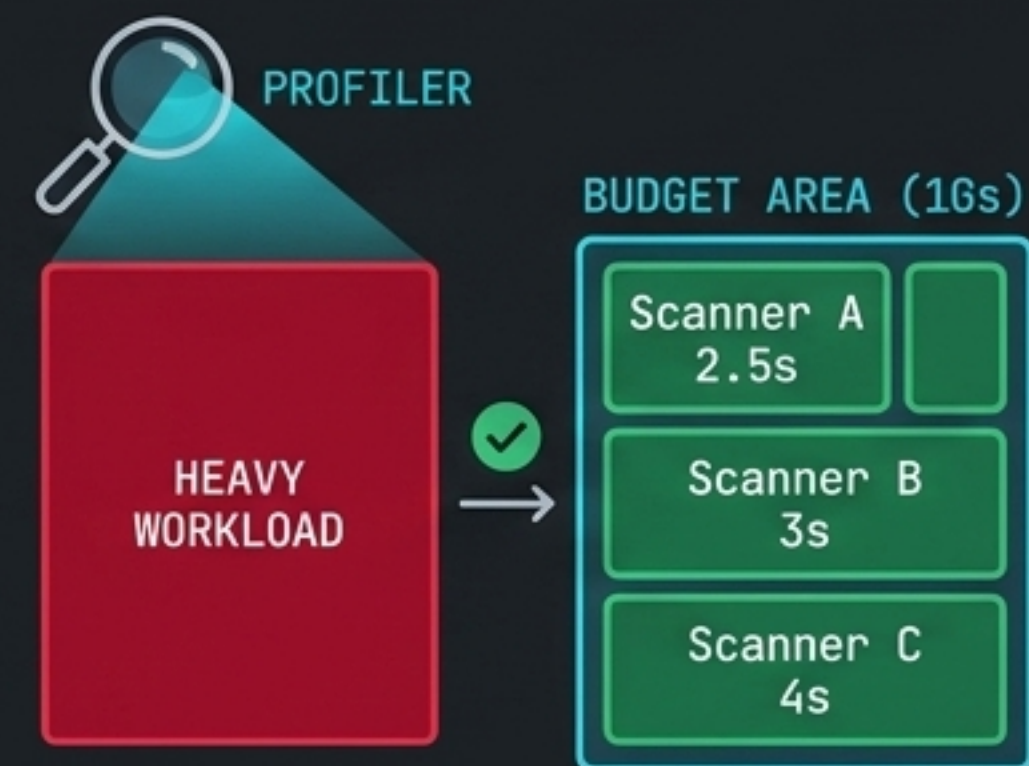
MECHANISM

Based on 202,229 real timing records,
the hook averages 19.1s with a p95 of
50.1s. It runs on the Stop event,
stalling the session stalling the
session exactly when the user is
waiting. The 10s timeout acts as
documentation, not enforcement.



AFTER

Profile which internal scanner
dominates. Fix the work to fit the
budget, or raise the budget and move
heavy work off the Stop path.



Priority 4: Unmatchable Deny Rules

2 active deny rules | 0% actual block rate

BEFORE	MECHANISM	AFTER
Two active deny rules attempt to block mass deletion: Bash(find:*-delete*) Bash(find:*-fprint*)	Permission rules match against the literal command string. No real invocation begins with find:. The command find . -delete is completely unblocked. These are silent no-ops presenting as protection. find:*-delete* find . -name '*.tmp' -delete SYNTAX MISMATCH: Literal colon does not match space or command argument structure.	Rules must match realistic command shapes, or be explicitly deleted so the security posture is honest about what it does not cover. # Deny mass deletion commands using regex deny: - name: "Block dangerous find delete" command_regex: '^find\s+.*\s+-delete.*\$' - name: "Block dangerous find fprint" command_regex: '^find\s+.*\s+-fprint.*\$'

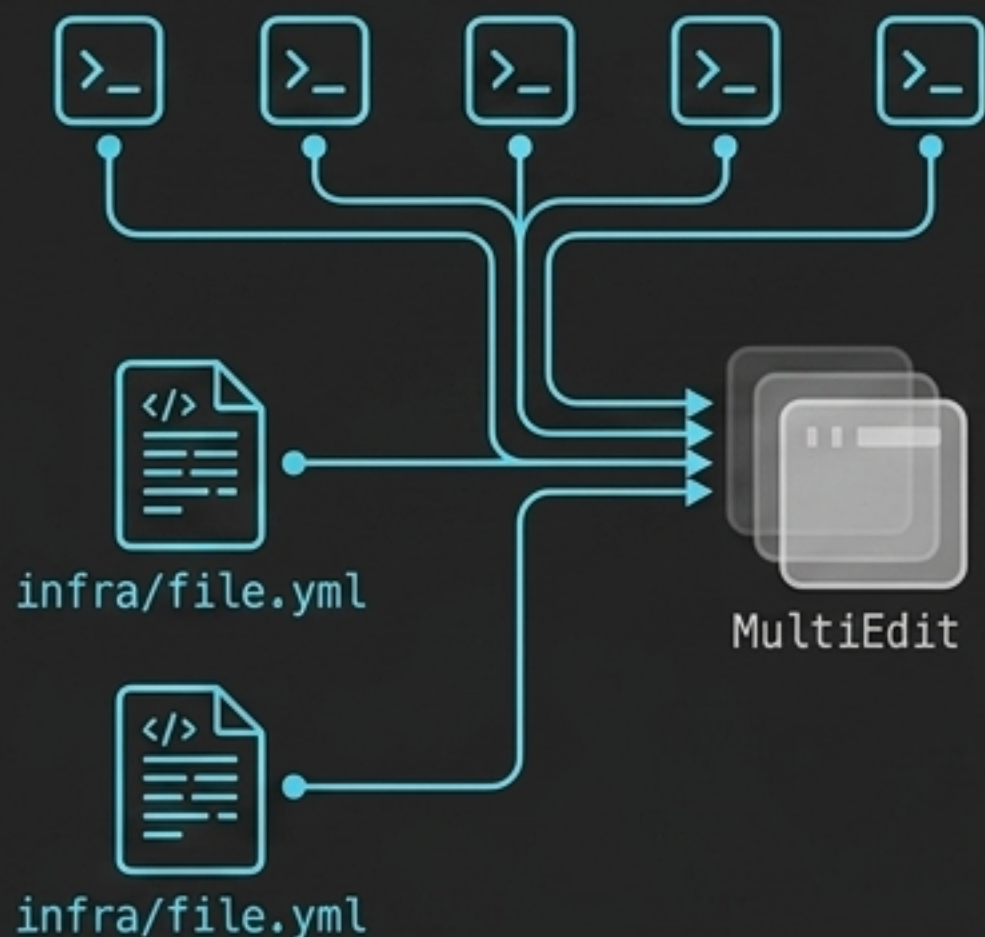
Priority 5: Missing Tool-Name Allowlist Validator

Telemetry:

15 existing validators | 0 tool-name checks | 5 agents impacted

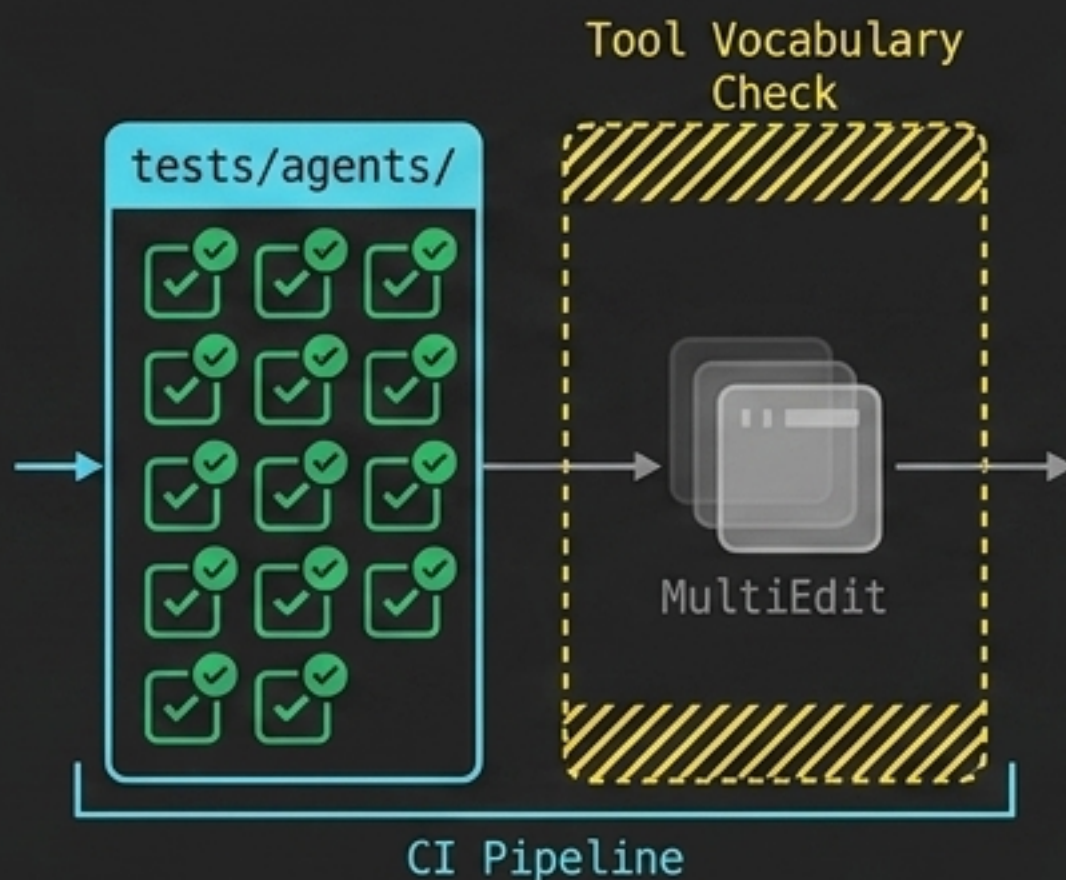
BEFORE

MultiEdit is granted by 5 agents and codified in 3 infra files, despite being dead and absent from live/deferred tool surfaces.



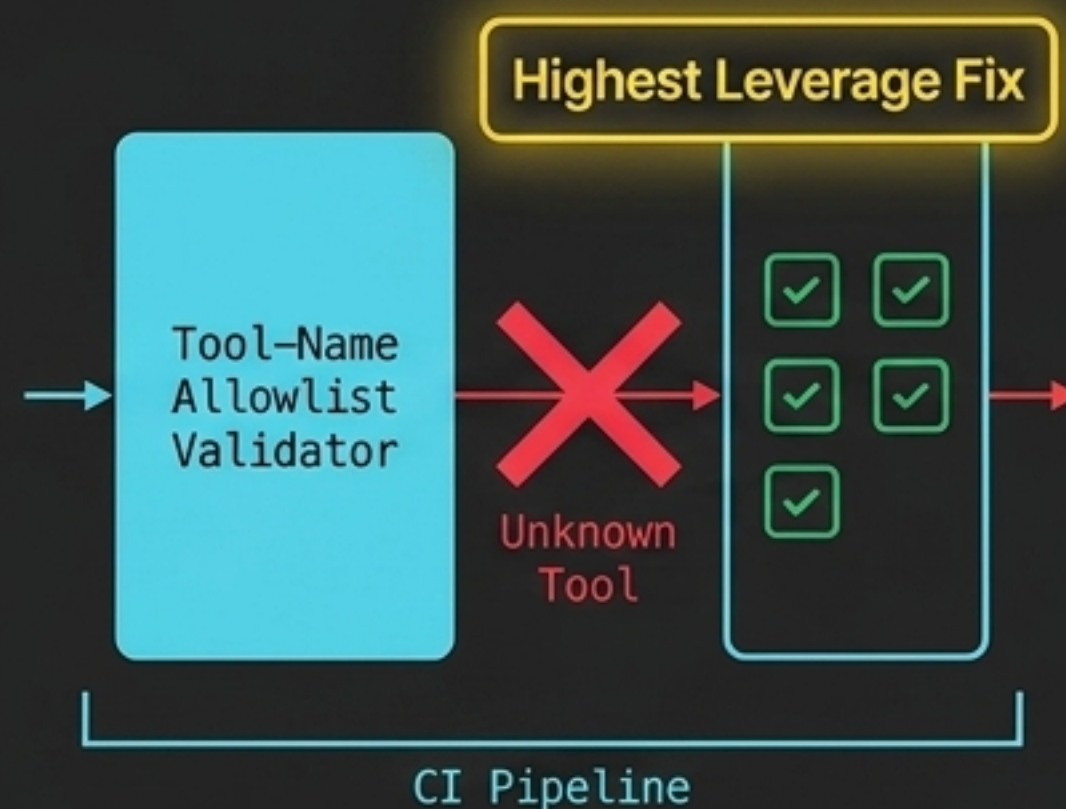
MECHANISM

15 existing validators in tests/agents/, yet 0 check declared tool names against a known vocabulary. The system cannot catch hallucinated or deprecated tool configurations.



AFTER

Implement a strict CI tool-name allowlist validator. Fails CI on any unknown tool. MultiEdit removed.



The Audit Boundary: What Did Not Survive Verification

Telemetry:
2 False Positives Identified & Discarded

REFUTED: *review-pr and explore define zero agent spawns.*

False. Initial sweep missed the rules/ directory. review-pr has 21 subagent spawns, explore has 22. Not a gap.

DOWNGRADED: *security-auditor instructs a spawn it cannot perform.*

False. Lines 85-92 instruct parallel Bash calls (npm audit, pip-audit), not agent spawns. No Agent grant required.

Architectural Mandates for the New Orchestration Model

1. Declare execution intent:

Every fork skill states **background: true** or **false**. Do not rely on scheduler defaults.

2. Spawn depth is a budget:

Own the cost explicitly via **depth variables**, or **fan out flat**.

3. Effort tiers the fan-out:

Heavy skills must read **CLAUDE_EFFORT** and scale agent count, not just prose.

4. Background implies a merge contract:

No agent may mutate files in a **backgrounded isolated worktree** unless the skill explicitly **merges the branch back**.